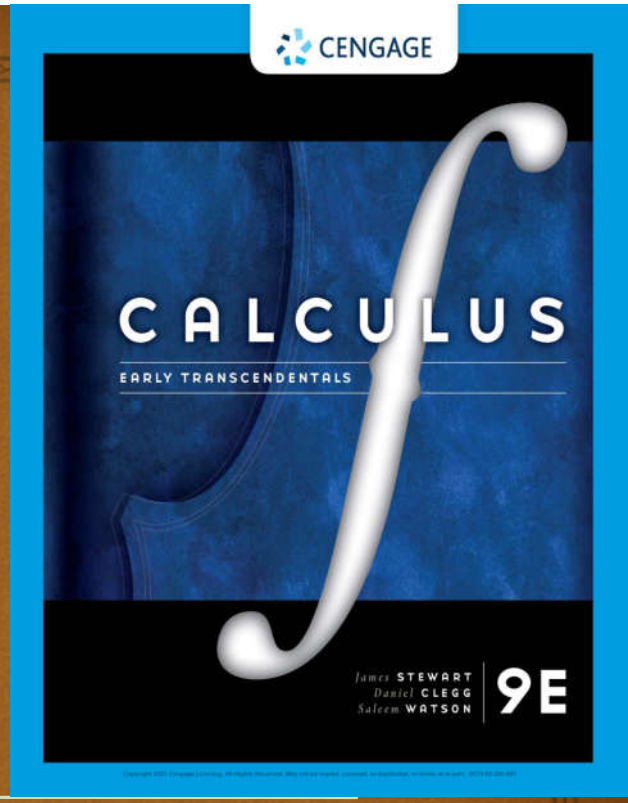
Chapter 1. SERIES



1.1 Sequences

p.724

Chapter 1

By the end of this lesson, you will be able to:

+ Understand about sequences,

+ Some properties of a sequence,

and limits of sequences.

I. Sequences

- **Def 1:** A **sequence** is a list of numbers written in a definite order:

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

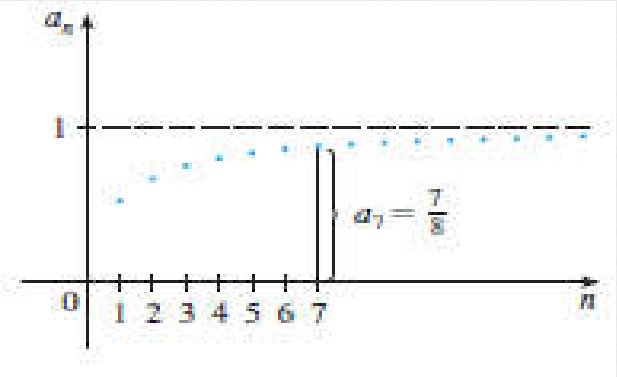
a_n is called the *nth* term of the sequence.

Notation: $\{a_n\}$ or $\{a_n\}_{n=1}^{\infty}$

- A sequence can be defined as a function whose domain is the set of positive integers (for every positive integer n there is a corresponding number a_n).

Sequences

Example 1. Some sequences can be defined by giving a formula for the *nth* term.



- $a_n = \frac{n}{n+1}$

  The sequence is $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}, \dots\right\}$

- $a_n = \frac{(-1)^n(n+1)}{3^n}$

  The sequence is $\left\{-\frac{2}{3}, \frac{3}{9}, -\frac{4}{27}, \frac{5}{81}, \dots, \frac{(-1)^n(n+1)}{3^n}, \dots\right\}$

- $a_n = \sqrt{n-3}, n \geq 3$

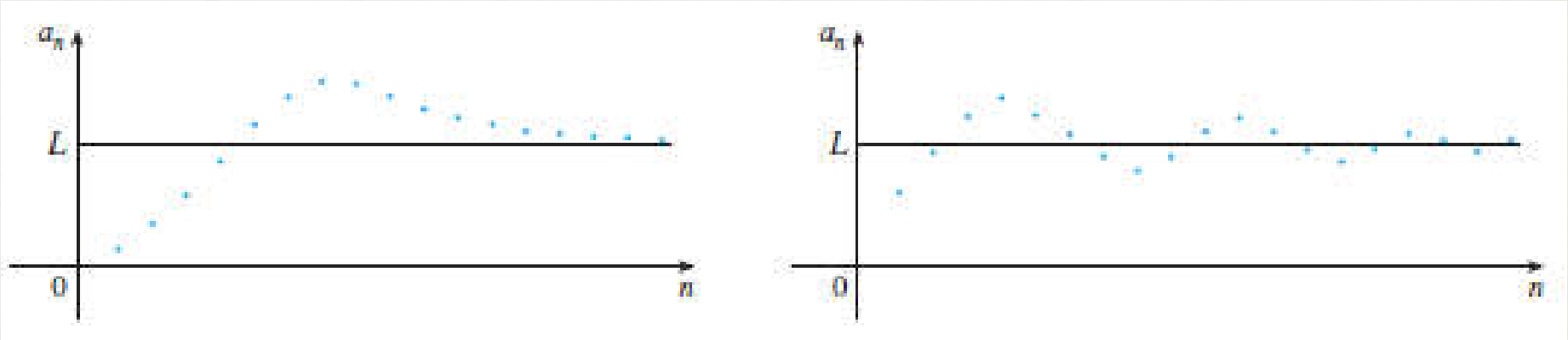
  The sequence is $\{0, 1, \sqrt{2}, \sqrt{3}, \dots, \sqrt{n-3}, \dots\}$

II. Limit of Sequence

1 DEFINITION A sequence $\{a_n\}$ has the **limit** L and we write

$$\lim_{n \to \infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

if we can make the terms a_n as close to L as we like by taking n sufficiently large. If $\lim_{n \to \infty} a_n$ exists, we say the sequence **converges** (or is **convergent**). Otherwise, we say the sequence **diverges** (or is **divergent**).



Sequences

2 Precise Definition of a Limit of a Sequence A sequence $\{a_n\}$ has the **limit** L and we write

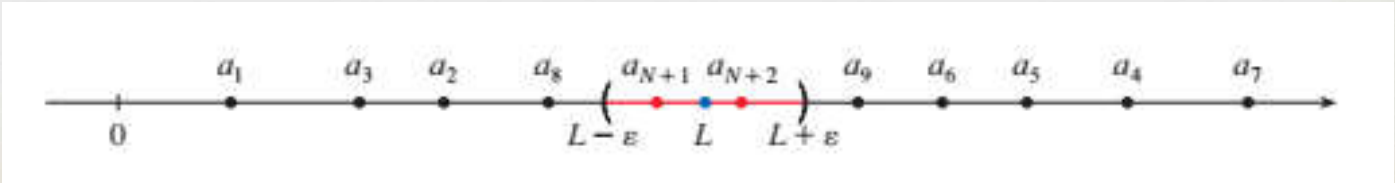
$$\lim_{n \to \infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

if for every $\varepsilon > 0$ there is a corresponding integer N such that

$$\text{if} \qquad n > N \qquad \text{then} \qquad |a_n - L| < \varepsilon$$

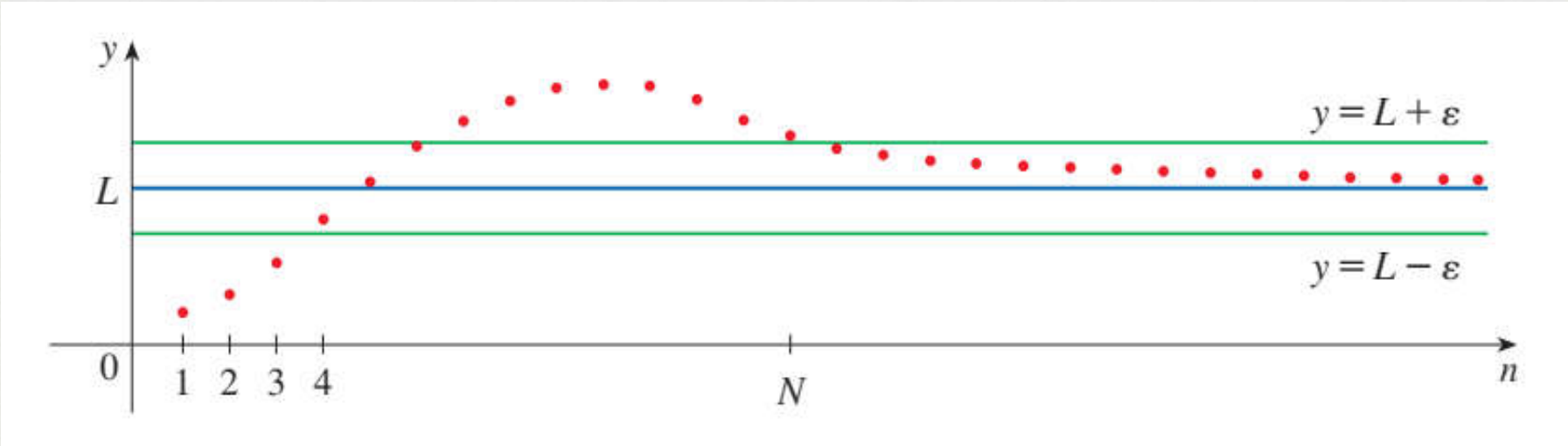
No matter how small an interval $(L - \varepsilon, L + \varepsilon)$ is chosen, there exists an N such that all terms of the sequence from a_{N+1} onward must lie in that interval.

$$a_1, a_2, a_3, \ldots, a_N, \underbrace{a_{N+1}, a_{N+2}, a_{N+3}, \ldots}$$



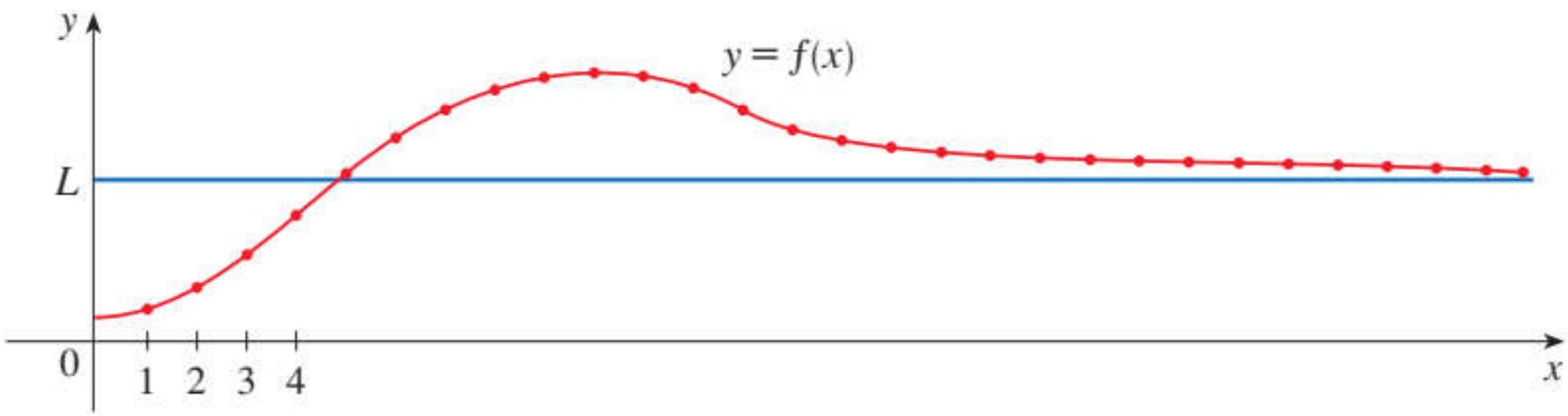
Sequences

- The points on the graph of the function $f(n) = a_n$ must lie between the horizontal lines $y = L - \varepsilon$ and $y = L + \varepsilon$ if $n > N$.



Sequences

4 Theorem If $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then $\lim_{n\to\infty} a_n = L$.

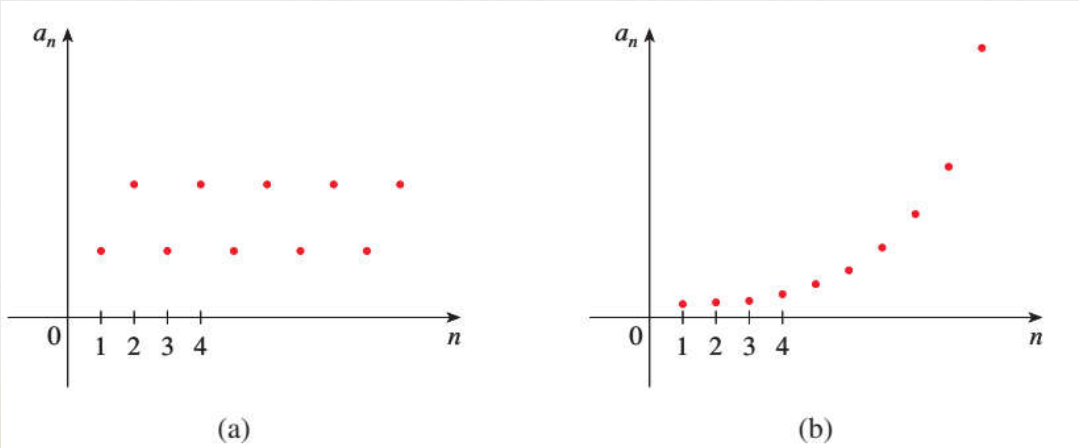


Sequences

5 DEFINITION $\lim_{n\to\infty} a_n = \infty$ means that for every positive number M there is a positive integer N such that

$$\text{if} \quad n > N \quad \text{then} \quad a_n > M$$

- The sequence $\{a_n\}$ is divergent or the sequence $\{a_n\}$ diverges to ∞.



Sequences

- Limit laws for sequences

If $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant, then

$$\lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty}(a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n$$

$$\lim_{n\to\infty} ca_n = c\lim_{n\to\infty} a_n \qquad \lim_{n\to\infty} c = c$$

$$\lim_{n\to\infty}(a_n b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$$

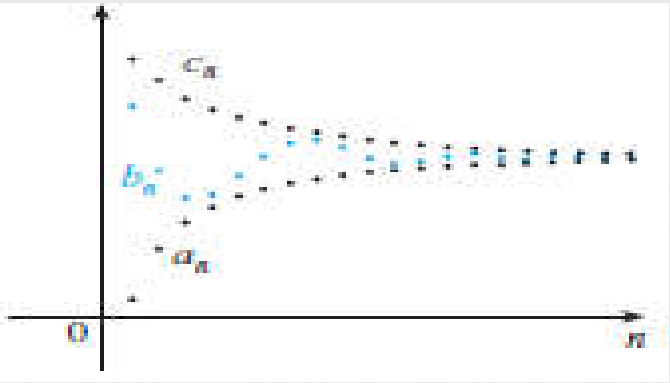
$$\lim_{n\to\infty}\frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} \quad \text{if } \lim_{n\to\infty} b_n \neq 0$$

$$\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p \quad \text{if } p > 0 \text{ and } a_n > 0$$

Sequences

- Squeeze theorem for sequences

If $a_n \leqslant b_n \leqslant c_n$ for $n \geqslant n_0$ and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.



The sequence $\{b_n\}$ is squeezed between the sequences $\{a_n\}$ and $\{c_n\}$.

6 THEOREM If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Sequences

Example 2. Find $\lim_{n\to\infty} \frac{\ln n}{n}$.

Consider the function $f(x) = \frac{\ln x}{x}$, we have

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\ln x}{x} \qquad \text{(by applying the L'Hospital's Rule)}$$

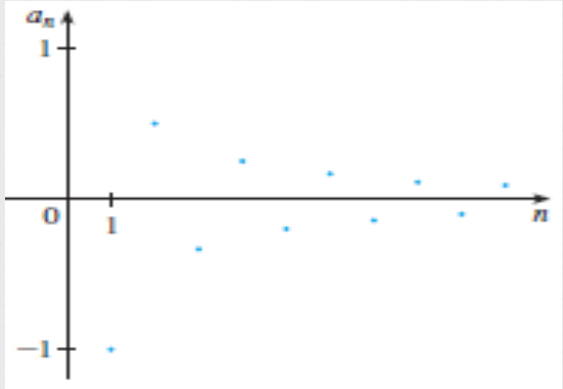
$$= \lim_{x\to\infty} \frac{1/x}{1} = 0.$$

Therefore, by the theorem 3, we obtain

$$\lim_{n\to\infty} \frac{\ln n}{n} = 0$$

Sequences

Example 3. Evaluate $\lim_{n\to\infty} \frac{(-1)^n}{n}$ if it exists.



Consider $\lim_{n\to\infty} \left|\frac{(-1)^n}{n}\right| = \lim_{n\to\infty} \frac{1}{n} = 0.$

Therefore, by theorem 6, we have

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

CONTINUITY AND CONVERGENCE THEOREM If $\lim_{n\to\infty} a_n = L$ and the function f is continuous at L, then

$$\lim_{n\to\infty} f(a_n) = f(L)$$

Sequences

Example 4. Find $\lim_{n\to\infty} \sin(\frac{\pi}{n})$.

Because $\lim_{n\to\infty} \frac{\pi}{n} = 0$ and the sin function is continuous at 0, we have

$$\lim_{n\to\infty} \sin(\frac{\pi}{n}) = \sin 0 = 0.$$

Example 5. Determine whether the sequence $a_n = \frac{n!}{n^n}$ converges or diverges.

We have $0 < a_n = \frac{1}{n} \times \frac{2.3.4....n}{n.n.n...n} < \frac{1}{n}$ and $\lim_{n\to\infty} \frac{1}{n} = 0$. Therefore, by Squeeze theorem, we conclude that $\lim_{n\to\infty} a_n = 0$ and the sequence converges.

Example 6. For what values of r is the sequence $\{r^n\}$ convergent?

SOLUTION We know from Section 2.6 and the graphs of the exponential functions in Section 1.4 that $\lim_{x\to\infty} b^x = \infty$ for $b > 1$ and $\lim_{x\to\infty} b^x = 0$ for $0 < b < 1$. Therefore, putting $b = r$ and using Theorem 4, we have

$$\lim_{n\to\infty} r^n = \begin{cases} \infty & \text{if } r > 1 \\ 0 & \text{if } 0 < r < 1 \end{cases}$$

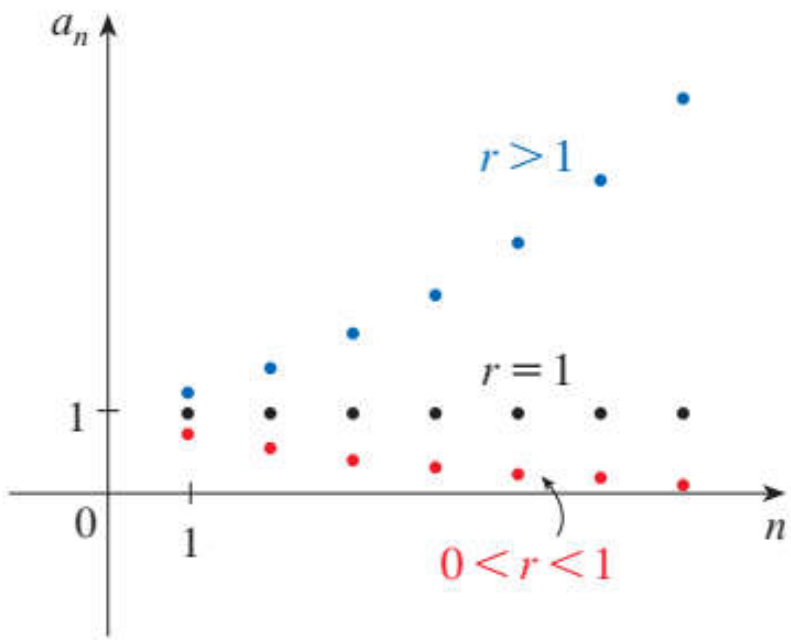
It is obvious that

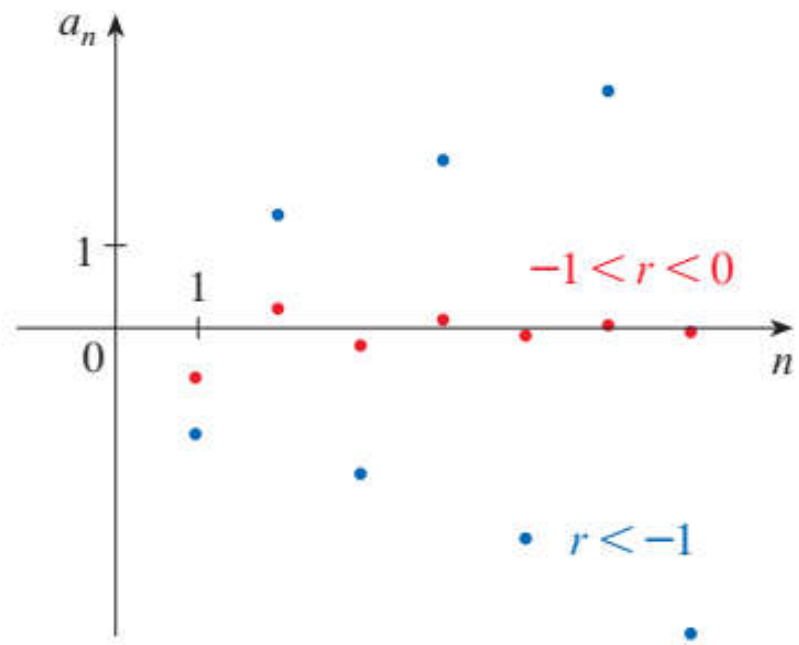
$$\lim_{n\to\infty} 1^n = 1 \qquad \text{and} \qquad \lim_{n\to\infty} 0^n = 0$$

If $-1 < r < 0$, then $0 < |r| < 1$, so

$$\lim_{n\to\infty} |r^n| = \lim_{n\to\infty} |r|^n = 0$$

and therefore $\lim_{n\to\infty} r^n = 0$ by Theorem 6. If $r \leqslant -1$, then $\{r^n\}$ diverges as in Example 7. Figure 13 shows the graphs for various values of r. (The case $r = -1$ is shown in Figure 10.)





9 The sequence $\{r^n\}$ is convergent if $-1 < r \leqslant 1$ and divergent for all other values of r.

$$\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$

III. Monotonic and Bounded Sequences

10 Definition A sequence $\{a_n\}$ is called **increasing** if $a_n < a_{n+1}$ for all $n \geqslant 1$, that is, $a_1 < a_2 < a_3 < \cdots$. It is called **decreasing** if $a_n > a_{n+1}$ for all $n \geqslant 1$. A sequence is called **monotonic** if it is either increasing or decreasing.

Example 7. Show that the sequence $\{\frac{3}{n+5}\}$ is decreasing.

Example 8. Show that the sequence $\{a_n = n^3 - 3n + 3\}_{n\geq 1}$ is increasing.

Sequences

Example 9. Show that the sequence $a_n = \frac{n}{n^2+1}$ is decreasing.

Solution:

Consider the function $f(x) = \frac{x}{x^2+1}$ $(x \geq 1)$.

We have $f'(x) = \frac{1-x^2}{(x^2+1)^2} < 0$ for $x > 1$. Thus, $f(x)$ is decreasing and so $f(n) = a_n$ is decreasing.

Sequences

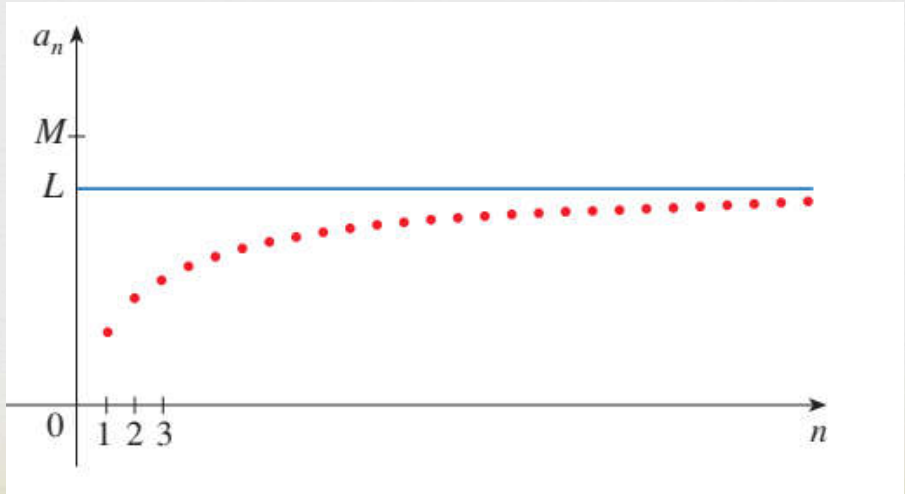
11 Definition A sequence $\{a_n\}$ is **bounded above** if there is a number M such that

$$a_n \leqslant M \qquad \text{for all } n \geqslant 1$$

A sequence is **bounded below** if there is a number m such that

$$m \leqslant a_n \qquad \text{for all } n \geqslant 1$$

If a sequence is bounded above and below, then it is called a **bounded sequence**.



Sequences

Example 10. The sequence $a_n = \frac{n}{n+1}$ is bounded because $0 < a_n < 1$ for all n.

12 Monotonic Sequence Theorem Every bounded, monotonic sequence is convergent.

In particular, a sequence that is increasing and bounded above converges, and a sequence that is decreasing and bounded below converges.

BASIC TERMINOLOGIES

1. Sequence
2. Limit of a sequence
3. Convergence and divergence of a sequence
4. Bounded sequence
5. Monotonic sequences: Increasing and decreasing sequences

EXERCISES

- **Exercises: Ex 8 – 16, 18, 19, 23-26, 29, 30, 34, 36, 42, 43, 44, 48, 53, 55, 56, 71, 72, 85, 96, 97 (P 736 - 737).**

1.2 Series

p. 739

Series

Given an infinite sequence $\{a_n\}$, an expression of the form

$$a_1 + a_2 + \cdots + a_n + \cdots$$

is called an **infinite series** (or just a **series**) and denoted by the symbol

$$\sum_{n=1}^{\infty} a_n \qquad \text{or} \qquad \sum a_n.$$

Example: Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots$$

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

The table shows that the sum of first n terms becomes closer and closer to 1 as n increases.

Series

2 DEFINITION Given a series $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots$, let s_n denote its nth partial sum:

$$s_n = \sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$$

If the sequence $\{s_n\}$ is convergent and $\lim_{n\to\infty} s_n = s$ exists as a real number, then the series $\sum a_n$ is called **convergent** and we write

$$a_1 + a_2 + \cdots + a_n + \cdots = s \qquad \text{or} \qquad \sum_{n=1}^{\infty} a_n = s$$

The number s is called the **sum** of the series. If the sequence $\{s_n\}$ is divergent, then the series is called **divergent**.

Notice that

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} \sum_{i=1}^{n} a_i$$

Series

Example 1. Suppose we know that the sum of the first n terms of the series $\sum_{n=1}^{\infty} a_n$ is

$$s_n = a_1 + a_2 + \cdots + a_n = \frac{2n}{3n+5}$$

Then the sum of the series is the limit of the sequence $\{s_n\}$:

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \frac{2n}{3n+5} = \frac{2}{3}$$

Series

Example 2. Show that the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ is convergent and find its sum.

Series

Example 3. Consider the sequence that each term is obtained from the preceding one by multiplying it by the common ratio r $(a_1 = a_0 r^0, a_n = ar^{n-1})$. The series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1} + \ldots$$

is called the geometric series.

The partial sums $s_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}$.

If $r = 1$, then $s_n = a + a + a + \cdots + a = na \to \pm\infty$.

Thus, the geometric series diverges.

Series

If $r \neq 1$, we have

$$s_n = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}$$
$$rs_n = ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n$$

Subtracting these equations, we get

$$s_n - rs_n = a - ar^n$$
$$s_n = \frac{a(1 - r^n)}{1 - r}$$

If $|r| < 1$, then $r^n \to 0$ as $n \to \infty$.

If $r \leq -1$ or $r > 1$, the sequence $\{r^n\}$ is divergent. Thus, the geometric series diverges.

Series

4 The geometric series

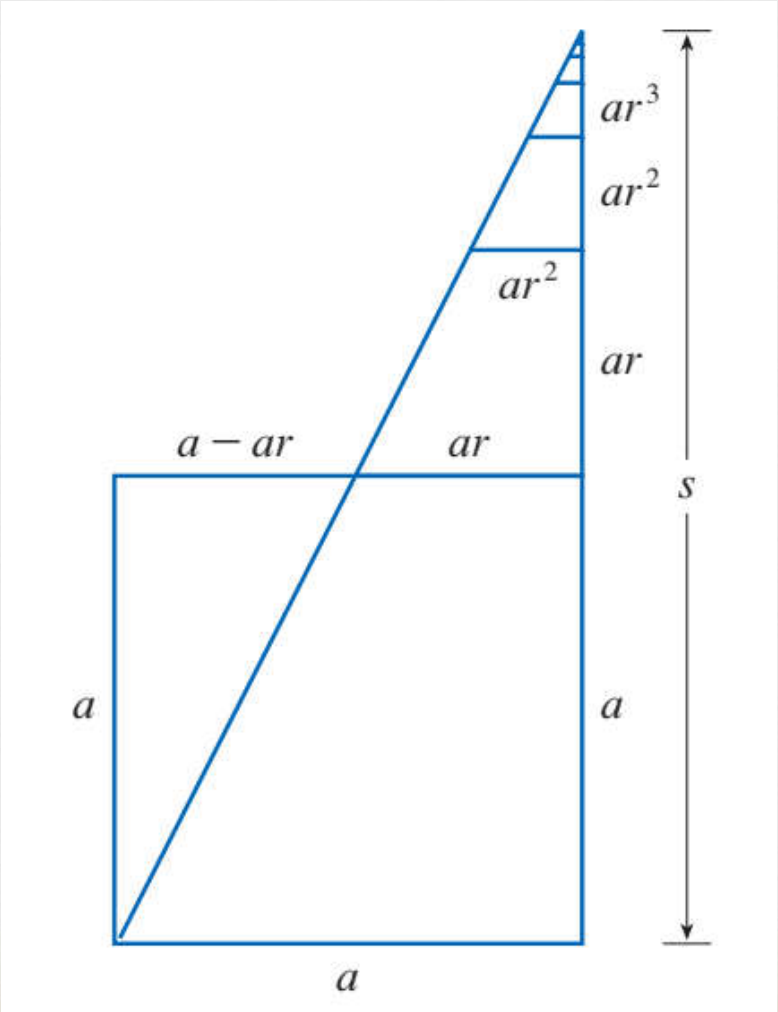
$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots$$

is convergent if $|r| < 1$ and its sum is

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \qquad |r| < 1$$

If $|r| \geqslant 1$, the geometric series is divergent.

Geometric series



Series

Example 4. Find the sum of the geometric series

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$$

The first term is $a = 5$, the common ratio is $r = -\frac{2}{3}$ and $|r| < 1$. Thus, the geometric series is convergent, and its sum equals to $\frac{5}{1-(-2/3)} = 3$

Example 5. Write the number $2.3\overline{17} = 2.3171717\ldots$ as a ratio of integers.

$$2.3171717\ldots = 2.3 + \frac{17}{10^3} + \frac{17}{10^5} + \cdots = \frac{23}{10} + \frac{\frac{17}{10^3}}{1 - \frac{1}{10^2}} = \frac{1147}{495}$$

Series

Example 6. Find the sum of the series $\sum_{n=0}^{\infty} x^n$, where $|x| < 1$.

SOLUTION Notice that this series starts with $n = 0$ and so the first term is $x^0 = 1$. (With series, we adopt the convention that $x^0 = 1$ even when $x = 0$.) Thus

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \cdots$$

This is a geometric series with $a = 1$ and $r = x$. Since $|r| = |x| < 1$, it converges and (4) gives

$$\boxed{5} \qquad \sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$$

Series

Example 7. Is the series $\sum_{n=1}^{\infty} 2^{2n}3^{1-n}$ convergent or divergent?

$$\sum_{n=1}^{\infty} 2^{2n}3^{1-n} = \sum_{n=1}^{\infty} \frac{4^n}{3^{n-1}} = \sum_{n=1}^{\infty} 4\left(\frac{4}{3}\right)^{n-1}$$

Thus, $a = 4$ and $r = \frac{4}{3} > 1$. So, the series is divergent.

Series

Example 8. Is the harmonic series $\sum_{n=1}^{\infty} 1/n$ convergent or divergent?

Suppose the series converges to H, we have

$$H = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \cdots$$

$$\geq 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} + \frac{1}{6} + \frac{1}{8} + \frac{1}{8} + \cdots$$

$$H \geq 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \frac{1}{2} + H$$

This contradiction concludes that the harmonic diverges.

Series

6 THEOREM If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$.

7 TEST FOR DIVERGENCE If $\lim_{n\to\infty} a_n$ does not exist or if $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Example 9. Show that the series $\sum_{n=1}^{\infty} \frac{n^2}{5n^2+4}$ diverges

$$\lim_{n\to\infty} \frac{n^2}{5n^2+4} = 1/5 \neq 0$$

Series

8 THEOREM If $\Sigma\, a_n$ and $\Sigma\, b_n$ are convergent series, then so are the series $\Sigma\, ca_n$ (where c is a constant), $\Sigma\,(a_n + b_n)$, and $\Sigma\,(a_n - b_n)$, and

(i) $\displaystyle\sum_{n=1}^{\infty} ca_n = c\sum_{n=1}^{\infty} a_n$

(ii) $\displaystyle\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$

(iii) $\displaystyle\sum_{n=1}^{\infty} (a_n - b_n) = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$

Series

Example 10. Find the sum of the series

$$\sum_{n=1}^{+\infty}\left(\frac{3}{n(n+1)}+\frac{1}{2^n}\right).$$

Series

NOTE 4 A finite number of terms doesn't affect the convergence or divergence of a series. For instance, suppose that we were able to show that the series

$$\sum_{n=4}^{\infty} \frac{n}{n^3+1}$$

is convergent. Since

$$\sum_{n=1}^{\infty} \frac{n}{n^3+1} = \frac{1}{2} + \frac{2}{9} + \frac{3}{28} + \sum_{n=4}^{\infty} \frac{n}{n^3+1}$$

it follows that the entire series $\sum_{n=1}^{\infty} n/(n^3+1)$ is convergent. Similarly, if it is known that the series $\sum_{n=N+1}^{\infty} a_n$ converges, then the full series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{N} a_n + \sum_{n=N+1}^{\infty} a_n$$

is also convergent.

BASIC TERMINOLOGIES

1. Series
2. Infinite series
3. Partial series
4. Convergent and divergent
5. Geometric series

EXERCISES

- **Exercises: Ex 8, 9, 10, 13, 14, 15, 17-22, 29-32 , 37 – 45, 48, 50, 62, 63, 64, 65, 86, 92, 95 (P 747 - 751)**